FLEET TEAM DIGEST

737MAX-FTD-32-20001

Issue Title : 737MAX Nose Landing Gear Orifice Plate Retainer Nut Coming Loose

Airplane Model	ATA	Minor Model(s)
737MAX	3221-00	-7, -8, -8200, -9, -BBJ
Other Model(s)	**ECCN**	
	9E991	

Originated Date	Last Revised Date	Created On
08/04/2020	08/04/2020	02/25/2020
Estimated Completion Date	**Next Update Date**	

Applicability

737MAX line numbers: 5602 - 8129.

Note: There are 737NG line numbers within these lines, refer to FTD article 737NG-FTD-32-11003.

Description

Note: The 737NG and 737MAX nose landing gear are very similar in design; hence, Boeing is releasing this FTD article for the 737MAX fleet. During 737MAX operations, Boeing has not received any similar cases for the 737MAX nose landing gear configurations.

Boeing has received 15 reports of 737NG nose landing orifice plate retainer nut, P/N 162A1524-1, has come loose during operation. The retaining nut can back-off and contact internal shock strut components if the lockwire elongates or fails. The lockwire and retaining nut will be out of position within the shock strut, allowing internal contact/damage. This will allow metal debris to intermix with the shock strut fluid, causing the nose landing gear to leak. In-service figures are provided within this FTD article for reference; please see attached file 'Reported In-Service Findings'.

Background

Boeing has received an increased number of 737NG reports where the nose landing gear shock strut was found leaking, and metal particle debris was found within the seals.

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Closer examinations of these events found:

- The single lockwire elongated but intact, with the orifice plate retaining nut out of position, and internal contact damage.
- The single lockwire fractured, and the orifice plate retaining nut out of position with internal contact damage. The orifice support tube threads may become damaged due to the mating orifice plate retainer nut being out of position.
- The single lockwire fractured and orifice plate retaining nut completely backed-off, falling from the shock strut during disassembly. The orifice plate retaining nut was found damaged along with the mating orifice support tube threads.

These conditions allow metal debris to intermix with the shock strut fluid, affecting the seals and causing the nose landing gear to leak. There were occasions were this anomaly was not discovered until the nose landing gear was being disassembled during scheduled overhaul.

Boeing has identified the following possible factors related to this condition:

1. The pigtail of the lockwire installed on orifice plate retainer nut, P/N 162A1524-1, extends past the face of the nut and comes into contact with metering pin retainer nut, P/N 162A1505-1. This can cause “premature fracturing of the lock wire”, allowing the Retainer Nut, P/N 162A1524-1, to rotate out of position and creating debris within the NLG shock strut.
2. A torque value of 30 -50 in-lbs, which was the previous requirement, and installation of one lockwire to the orifice plate retainer nut, P/N 162A1524-1, may not provide sufficient anti-rotation protection.

Status

Boeing actions complete - see Final Action.

Interim Action

If you are experiencing nose landing gear shock strut leakage, examine the removed seals and shock strut fluid for signs of metal debris. If debris is found, Boeing recommends performing a detail visual inspection, using a mirror with an extension arm and adequate light source, to inspect the lockwire and orifice plate retaining nut.

- If damage is found, Boeing recommends disassembling the shock strut per CMM 32-21-12 to evaluate the extent of the internal hardware damage. This may need to be performed in a shop environment.

- If no damage is found to the orifice plate retaining nut, but the lockwire has fractured, it is acceptable to disassemble the NLG shock strut per CMM 32-21-12 and replace the lockwire. This may need to be performed in a shop environment.

Final Action

Boeing CMM 32-21-12, dated 01-Jul-2020, revised the orifice plate retaining nut torque; it was increased from 30 - 50 in-lbs to 25 - 40 ft-lbs.

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Engineering drawings have been revised to require all orifice plate retaining nut to be installed with four lockwire in lieu of one.

Engineering drawings have been revised to include a new view showing details of how the lockwire pigtail is to be tucked-in to avoid any possible contact when the shock strut is compressed during in-service operations.

Milestones

Change Committed: 24-Feb-2020

Production Incorporation: Line 8130

CMM 32-21-12 revision date 01-Jul-2020

Part Information

NLG Orifice Plate Retainer Nut P/N 162A1524-1

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Drawing	162A1100		
Component Maintenance Manual (CMM)	32-21-12	44	11/01/2014

Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
Orifice Plate Retaining Nut	162A1524-1	Goodrich Aerospace Canada LTD	186501

Related Categories

- AOGs : Events that cause a significant Airplane On Ground (AOG) condition
- Best Practices / Operational Efficiency : Operational / Maintenance-focused best practices, techniques, health checks or recommendations for preserving aircraft efficiency.
- Component Issues : Issues that occur during component maintenance or have an impact on component reliability
- Dispatch Reliability Issues : Issues that cause revenue flight delays and cancellations

Attachments:

a) Reported_In-Serice_Findings

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b) Damaged_Lockwire

c) New Lockwire Installation Requirements

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